Software Requirements Specification

Smart Elevator System

Integrated Dispatch Controller, Cabin and Control Center

Expanded, Research-Grounded Hybrid SRS | Version 1.1

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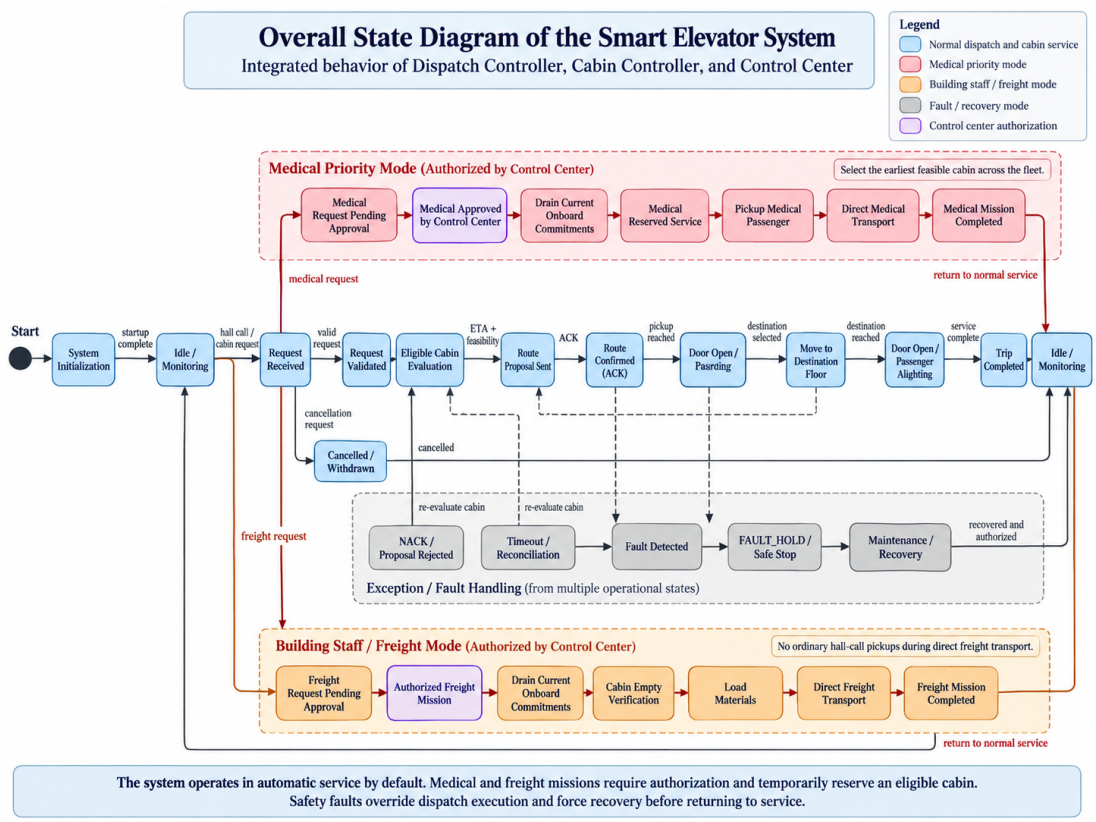


Figure 1: Whole-system integration: human interfaces, Dispatch Controller, Cabin Controller, Control Center, and simulated hardware.

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Revision History

Version	Change	Status
0.7	Original integrated SRS with three subsystems and proposed fleet policies.	Source draft
0.8.2	Consolidated Demo 2 format, route/ACK and cumulative-delay corrections.	Source draft
0.9	Reduced optional technologies and long test catalog; foregrounded algorithm, research and worked evidence.	Proposed submission
0.9.1	Concise SATSA research extension and gRPC/proto3 interface option; original mandatory dispatch and safety rules unchanged.	Team / instructor review
1.0	SATSA specified as a System Feature with equations, constraints, approval and verification; gRPC target clarified.	Submission design draft
1.1	Expanded hybrid SRS: state machines, simulator/event contracts, Agile stories, traceability and design-stage acceptance.	Team / instructor review

Reading route. Sections 1–3 cover scope and interfaces. Section 4 has the dispatch and SATSA algorithms, and Sections 5–6 cover requirements, research and feasibility. Section 7 describes the state machines and the event-driven simulation. Sections 8–9 tie the requirements to user stories, acceptance evidence and a release plan.

1 Introduction

1.1 Purpose

This SRS describes a software-only, event-driven Smart Elevator System for a 40-floor building. It covers how the Dispatch Controller, the Cabin Subsystems and the Control Center work together, from a user pressing a button through route assignment, simulated movement, monitoring and exception handling. There are enough behavior rules, input/output contracts and calculations here for the team to build a consistent prototype, and for the instructor to judge whether the design is technically feasible.

1.2 Document Conventions

Requirement IDs are unique and grouped by who is responsible: DC (Dispatch), CB (Cabin), CC (Control Center), MED (Medical), FRT (Freight) and SAF (Safety). “Shall” marks required behavior within the proposed software baseline. Times

are in simulation seconds unless we say wall-clock time. Every worked calculation uses the same model: 2 s per floor of travel and a 6 s dwell at each served stop. Academic sources are cited as [n].

1.3 Intended Audience and Reading Suggestions

This document is for the instructor, developers, testers and team reviewers. If you're reading it for the first time, start with the architecture in Figure 1, the request-to-cabin sequence in Figure 2 and the worked scenarios in Sections 4.7–4.9. Then move on to the implementation details.

1.4 Product Scope

In scope: ordinary Hall Calls using UP/DOWN buttons; picking a Cabin Destination after boarding; safe assignment across passenger cars P1–P4; authorized MEDICAL pickups; Building Staff material transport; modeled door, load and fault controls; operator monitoring; and reproducible simulations. G1 is a separately configured goods car, and freight requests consider it first when it's eligible. The design also includes offline tuning of the dispatch grouping parameters with SATSA. That tuning stays offline: runtime movement control and certified hardware control are outside it. Out of scope: paid priority, premium travel, cameras that track passengers, real motor actuation, certified braking functions and hardware installation.

1.5 References

[1] IEEE, IEEE Std 830-1998: IEEE Recommended Practice for Software Requirements Specifications (format reference).

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[3] P. Cortés, J. Larrañeta and L. Onieva, "Genetic algorithm for controllers in elevator groups: analysis and simulation during lunchpeak traffic," Applied Soft Computing, 4(2), 159–174, 2004. https://doi.org/10.1016/j.asoc.2003.11.002

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[5] R. H. Crites and A. G. Barto, "Elevator group control using multiple reinforcement learning agents," Machine Learning, 33, 235–262, 1998. https://doi.org/10.1023/A:1007518724497

[6] Smart Elevator project team, Complete SRS v0.7 and integrated v0.8.2 (internal design drafts, October 2026).

[7] Instructor-provided [SRS] Demo 2 (VinUniClubs) sample, 2025 (structure and presentation reference only).

[8] J. Liu, Y. Hou, Y. Li and H. Zhou, "Advanced strategies on update mechanism of tree-seed algorithm for function optimization and engineering design problems," Expert Systems with Applications, vol. 236, art. 121312, 2024. https://doi.org/10.1016/j.eswa.2023.121312

[9] gRPC Authors, "Introduction to gRPC," gRPC Documentation. https://grpc.io/docs/what-is-grpc/introduction/ (accessed 18 October 2026).

[10] Instructor-provided Module 5 SRS Format Comparison sample (44-page teaching handout, 2026). Used for structure, user stories and traceability; not a technical source for elevator performance.

2 Overall Description

2.1 Product Perspective

The product has one authoritative Dispatch Controller, one independent Cabin Controller per elevator and one Control Center. Dispatch assigns service but can't bypass a Cabin Controller's local safety veto. Every message goes through a deterministic event model, so competing calls, acknowledgments and faults can be reproduced exactly.

Subsystem	Receives	Produces / owns
Dispatch Controller	Hall Calls; Cabin telemetry; approved missions	Canonical requests, candidate ETA, route proposal, single request owner
Cabin P1–P4 / G1	Proposed route; cabin floor input; sensor signals	ACK/NACK; doors/motion simulation; committed stops; telemetry
Control Center	Mission proposals; health/events; operator actions	Authorization, safety alerts, approval/recovery records

2.2 Product Functions

Normal trips: A passenger presses UP or DOWN. Dispatch evaluates the eligible cars and proposes a route. The Cabin confirms it can run that route safely and reports the pickup, and the destination is accepted only after boarding. Privileged missions: Control Center authorizes MEDICAL or FREIGHT, Dispatch finds an eligible car, the Cabin finishes the passengers already onboard, and then the car runs an exclusive route. Safety and recovery rules come before any scheduling priority. Offline parameter optimization: a simulation-based SATSA engine proposes versioned grouping settings for Control Center to review. It never generates motion commands.

2.3 User Classes and Characteristics

User	Permitted interaction
Passenger	UP/DOWN, in-cabin destination, alarm; no special mode approval
Medical Coordinator	Submit an authenticated medical transport mission
Building Staff	Declare material type, weight, pickup and destination
Control Center Operator / Supervisor	Approve missions; acknowledge faults; authorize return to service
Maintainer	Record simulated maintenance and fault corrections

2.4 Operating Environment

The baseline evaluation uses floors 1–40, passenger cabins P1–P4, goods cabin G1, a fixed 2 s of movement per floor and a 6 s dwell at each served stop. Movement, door state, occupancy and faults are all simulated through software adapters. The route decision service and each Cabin Controller can run as separate processes or modules. The contract doesn't depend on the networking technology.

2.5 Design and Implementation Constraints

Three things are hard constraints: Cabin safety interlocks, accepted onboard destinations and unique Hall Call ownership. The system shall not guess an unboarded passenger's destination from an ordinary UP/DOWN Hall Call, and it shall not deliberately hold a new ordinary request to build a group. The application shall not claim certification for physical elevators. The team still has to approve all numerical policy defaults.

2.6 Assumptions and Dependencies

In the controlled examples, positions, time and sensor readings are deterministic, though faults can be injected. An approved medical mission comes with an authorized pickup and delivery floor. An ordinary Hall Call gives only an origin and a direction. The building configures which cabins may carry the specified material or do medical transport.

3 External Interface Requirements

3.1 User Interfaces

Hall Panel: floor and UP/DOWN input, plus a visible accepted/waiting indicator. Cabin Panel: destination selection (permitted floors only), an alarm button and door/floor indicators. Control Center: fleet status, a form for privileged missions, alert review and return-to-service approval.

3.2 Hardware Interfaces (Simulated)

Cabin adapters report continuous cabin position, current direction of motion, door-zone confirmation, lock state, load, obstruction and fault flags. Simulated motion and door commands go only through the CabinController safety guards. Real motor control, emergency braking circuits and rescue operations are out of scope.

3.3 Software and Communications Interfaces

At a minimum, the implementation exchanges the logical messages below. Each one is designed to work in a gRPC/Protocol Buffers deployment (Section 3.4), and the early simulator can also run them in-process without changing the logical contracts. Route ownership, version checking and Cabin ACK/NACK behave the same either way.

Message	Essential data	Behavior
HallCall	call_id, origin, direction, timestamp	Create or return canonical active call
RouteProposal	command_id, car_id, expected_route_version, stops	Cabin validates; replies ACK/NACK
CabinTelemetry	car_id, time, position, direction, doors, load, mode, faults	Update eligibility and monitor health
MissionApproval	mission_id, type, origin, destination, authorization	Start eligible-medical/freight selection
ServiceEvent	call/mission id, stop, event, timestamp	Confirm pickup, delivery or fault

3.4 gRPC / Protocol Buffers Communication Profile

When the Dispatch Controller, Cabin Controllers and Control Center run as separate processes, the target design uses gRPC with Protocol Buffers (proto3) to exchange typed requests and telemetry [9]. We kept the service boundary small on purpose. Dispatch receives SubmitHallCall. Each Cabin receives ProposeRoute and returns a route ACK/NACK. Cabin telemetry streams to Dispatch and the Control Center. A single-process simulator can implement the same logical interfaces without any networking.

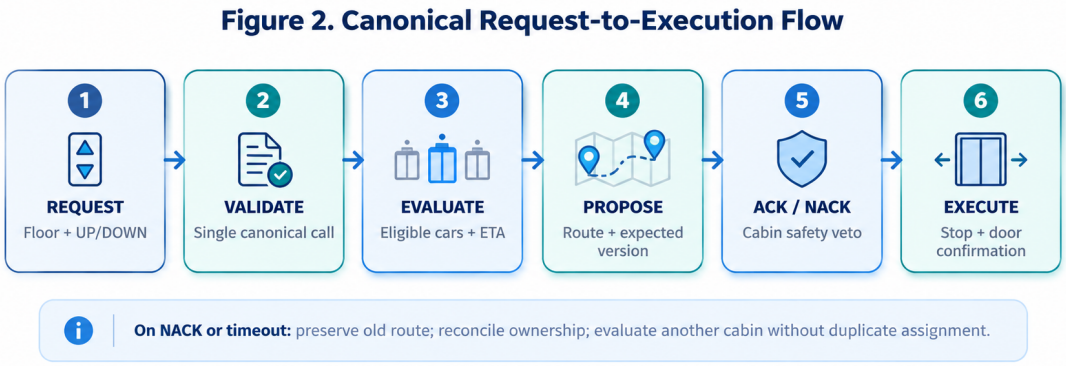


Figure 2: The canonical flow from request to execution. The addressed Cabin decides whether to accept a route, and an ambiguous acknowledgment never creates a second owner.

Illustrative payload fields: HallCall {event_id, origin_floor, direction, t_sim}; RouteProposal {command_id, car_id, expected_route_version, ordered_stops, expiry_ts}; RouteReply {ACK/NACK, accepted_version, reason}; Telemetry {car_id, position, direction, load, doors, safety_state, route_version}. These are minimal design contracts, not an implemented or compiled .proto file. If a Hall Call duplicates one that's already accepted, it gets the existing request_id back and no new request is created.

Communication rule. Every route command is idempotent by command_id. If an ACK is lost or a deadline expires, Dispatch keeps the request reserved. It then reconciles the Cabin's route version before it retries or releases ownership. A route that is valid but unsafe gets an application-level NACK, and a transport timeout is reported separately. Neither outcome allows a duplicate cabin assignment. Local Cabin safety checks stay authoritative whatever the transport.

4 System Features and Algorithms

4.1 Ordinary Passenger Service (High Priority)

Description. The system turns a hall button press into one active request, picks a feasible passenger cabin and keeps the commitments already made to onboard passengers. Stimulus/response: press 27 DOWN → canonical Hall Call → eligible car assignment → pickup at floor 27 → passenger selects floor 5 → committed cabin destination.

DC-01 Validate floors 1–40, forbid DOWN on floor 1 and UP on floor 40 and deduplicate active (floor, direction) requests.

DC-02 Evaluate eligible cars using current position, direction, load, mode, sensor freshness, existing stops and expected pickup ETA.

DC-03 Allow same-direction grouping only if floor span $\leq R = 3$ and fixed anchor time window $\leq W = 5$ s; make an immediate assignment attempt without waiting for the window to end.

DC-04 Protect accepted stops using a fixed commitment ETA reference; reject any insertion that increases an existing stop ETA by more than DELAY_CAP = 10 s cumulatively.

DC-05 Prefer earliest feasible pickup ETA; use stable tie-breaking, then compatible grouping where it does not breach existing commitments or materially disadvantage the anchor call.

DC-06 Protect a Hall Call aged $\geq$ AGE_LIMIT = 60 s by evaluating its earliest feasible assignment ahead of ordinary new work; this threshold is not a guaranteed 60-second pickup.

4.2 Cabin Execution and Local Safety (Critical)

Description. A Cabin Controller owns the simulated motion and doors, and it runs its own local feasibility checks no matter what score Dispatch gave the route.

CB-01 Accept a RouteProposal only if the route version, current position, door/lock state, load, operational mode and stopping constraints are valid; otherwise return a reasoned NACK and keep the previous route.

CB-02 Permit movement only with confirmed locked doors, load $\leq$ MAX_LOAD = 1000 kg and no motion-inhibiting fault. At FULL_LOAD = 900 kg, bypass new ordinary hall pickups but preserve onboard trips.

CB-03 Register a valid cabin destination after boarding and keep it committed until a verified destination stop or safety

exception.

CB-04 Complete a Hall Call upon a verified correct-direction pickup opportunity at its origin; complete a Cabin Destination upon verified drop-off at its own destination. These are distinct lifecycle events.

CB-05 On overload, hold doors and motion; permit identified passengers to unboard manually in the simulator and create any new Hall Call at the actual floor where that passenger exits.

4.3 Control Center and Special Mission Authorization (High Priority)

Description. Authorized operators control medical and staff privileges. A passenger alarm reports an incident, but it never quietly turns into a medical priority command.

CC-01 Check role, requested floors and mission type before accepting medical or material requests. Rejected grants shall not change Cabin route state.

CC-02 Display P1–P4/G1 mode, position, door status and faults. Report no-eligible-car conditions explicitly rather than showing false assignment success.

CC-03 Approve a mission only when its origin, destination and compatible cabin capabilities have been specified.

CC-04 Require a supervisor-mediated safe recovery step before a faulted cabin returns to automatic service.

4.4 MEDICAL: Fleet-Wide Earliest Feasible Pickup (Critical)

Description. MEDICAL is the highest-priority authorized mission for new work. It never overrides hard safety constraints or the unfinished trips of passengers already onboard. The dispatcher compares every eligible car instead of simply picking the one nearest by floor distance.

MED-01 Exclude faulted, stale, unauthorized or incompatible cars and estimate each remaining car's medical pickup time after its existing onboard passenger drop-offs and safe handoff.

MED-02 Select the lowest feasible medical pickup ETA, with deterministic car-ID tie breaking, and confirm the selection through Cabin ACK.

MED-03 Return ordinary unserved Hall Calls from the selected car to the active pool after safe ownership reconciliation, preserving original timestamps.

MED-04 If no car qualifies, retain the mission with an explicit UNAVAILABLE state, raise an urgent operator alert and reevaluate upon fleet-state change.

MED-05 After verified medical boarding, reserve the authorized medical destination as a direct mission leg without intermediate ordinary pickups; Cabin safety veto remains authoritative.

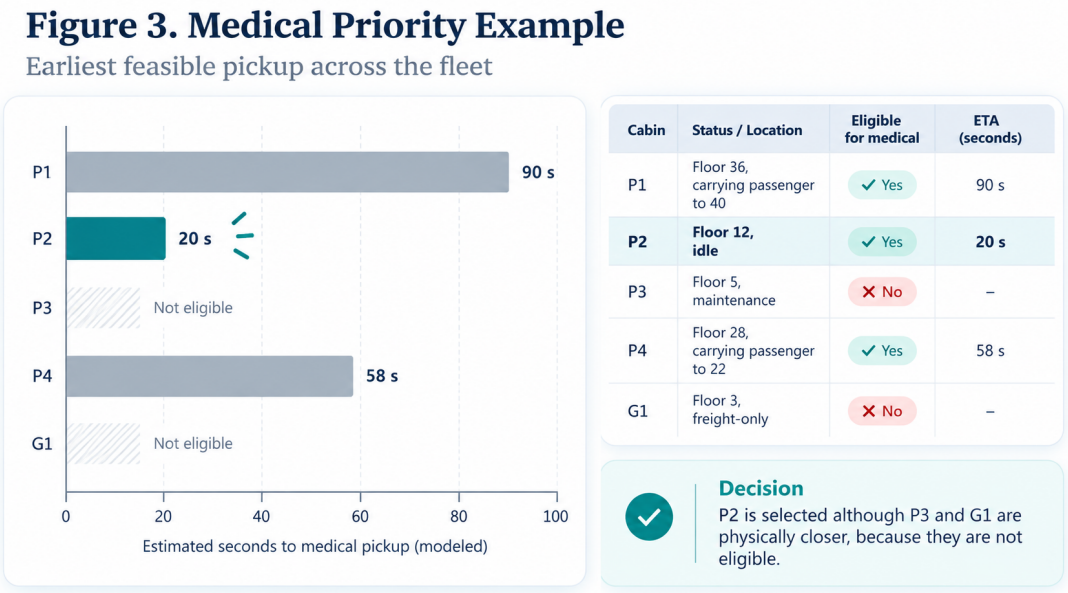


Figure 3: Medical call at floor 2. P2 is selected on modeled ETA rather than geometric closeness; P3 and G1 have no valid medical capability in this example.

4.5 Building Staff Material Transport (High Priority)

Description. Authorized Building Staff request a pickup floor, a destination, a material category and a load. G1 is preferred when it's eligible. Any configured passenger car can be used if policy allows it and it can finish its current passenger commitments first.

FRT-01 Authenticate staff and validate the material declaration, floors, rated payload and cabin capability before choosing a car.

FRT-02 Once reserved, the selected car finishes already onboard passenger trips and verifies it is stationary, empty and safe before moving to material pickup.

FRT-03 After staff confirms loading and Cabin load/door checks pass, run directly to the approved destination without ordinary intermediate Hall Calls. Safety overrides remain valid.

FRT-04 On delivery and unloading confirmation, release the exclusive reservation and restore normal service; failed loading or fault produces an explicit held/failed mission.

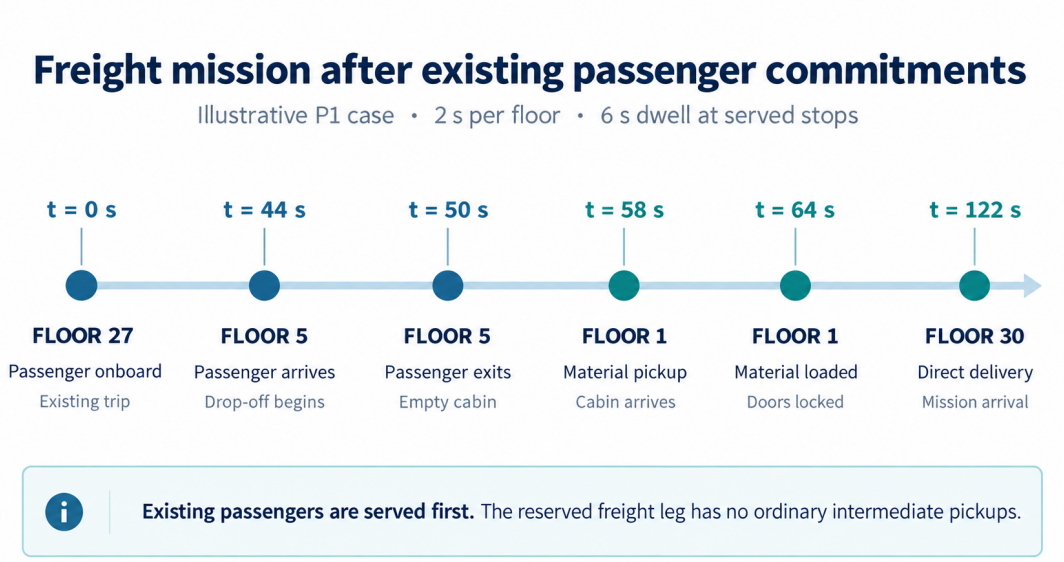


Figure 4: Freight scenario on P1: the currently onboard passenger is dropped off before materials are collected and delivered directly.

4.6 Safety, Fault Isolation and Recovery (Critical)

SAF-01 Never command normal movement with unlocked doors, invalid position, overload or a motion-inhibiting fault.

SAF-02 Do not open doors while moving or outside a valid landing door zone; reopen a closing door when an obstruction is detected.

SAF-03 On a modeled between-floor serious fault, stop simulated motion, hold cabin doors locked, alert Control Center and suspend onboard travel. Reassign unfinished external Hall Calls to other eligible cars only after ownership reconciliation.

SAF-04 Do not assign an active Hall Call to more than one car. On missing ACK, hold reservation until the last accepted route version is confirmed.

4.7 Dispatch Algorithm: Data, Equations and Decision Procedure

4.7.1 What the algorithm knows

An ordinary Hall Call is (origin, direction, created_at). Its destination stays unknown until someone selects a Cabin Destination after boarding. A mission approved by Control Center may include both origin and destination. The dispatcher reads ordered floors and committed stops, but it may never make up a passenger's destination.

4.7.2 Data structures

Structure	Core responsibility	Reason
HashMap / RequestRepository	Canonical call by ID and (origin, direction)	Detect duplicate active calls and maintain ownership
Ordered floor index (TreeMap or equivalent)	Hall Calls by floor and direction	Find nearest/next stops in a sweep efficiently
Deque / ordered stop list	Per-cabin committed route	Maintain stop order; evaluate insertions on a copied route

A Fenwick Tree could serve as an optional range-count index, but a 40-floor fleet doesn't need one. SATSA is planned as an offline parameter-optimization feature (Section 4.8). The live dispatch decision still relies on deterministic eligibility, ETA ranking, commitment limits and Cabin-local safety validation.

4.7.3 Equations and hard feasibility checks

Travel time and estimated stop arrival.

$$\text{TravelTime}(a, b) = 2\,|a - b| \text{ s}, \tag{1}$$

$$\text{ETA}(S_k) = t_{\text{current}} + \sum T_{\text{travel}} + \sum T_{\text{prior dwell (6 s/stop)}}. \tag{2}$$

Protection of previously committed stops.

$$\begin{aligned} \text{AddedDelay}(S_j) = {} & \text{ETA}_{\text{candidate}}(S_j) \\ & - \text{ETA}_{\text{committed reference}}(S_j) \le 10 \text{ s}. \end{aligned} \tag{3}$$

Same-direction grouping conditions.

$$\max(\text{floor}) - \min(\text{floor}) \le 3, \qquad \max(t) - \min(t) \le 5 \text{ s}. \tag{4}$$

Estimated MEDICAL pickup time.

$$\begin{aligned} \text{MedicalETA}(\text{car}) = {} & T_{\text{remaining onboard travel}} \\ & + T_{\text{required drop-off dwells}} \\ & + T_{\text{deadhead to mission origin}}. \end{aligned} \tag{5}$$

Committed-reference ETA values are fixed when a stop is assigned, and they are not reset after each new insertion. If they were reset, two +6 s insertions could each be accepted on their own, even though together they add 12 s and break the 10 s limit. When screening candidates, a moving car may accept a newly inserted stop only if at least one modeled floor remains ahead in its direction. That is a discrete simulator rule, not a real braking-distance formula.

4.7.4 Readable pseudocode

```
ON_HALL_CALL(call):
  c = validate_and_deduplicate(call)
  if c.already_active: return c.request_id
  reserve_pending(c)
  for car in eligible_passenger_cars():
    for candidate_route in safe_directional_insertions(car, c):
      if local_position_cutoff_fails(candidate_route): continue
      if any_committed_delay_over_10s(candidate_route): continue
      rank(candidate_route, pickup_ETA, grouping_tie_break)
  if no feasible route: leave c PENDING; log why; return
  choose best feasible route; reserve one owner
  propose_route_to_cabin(command_id, expected_version)
  if ACK: confirm assignment and preserve ETA references
  else: keep old route; reconcile; try next eligible car
ON_APPROVED_MEDICAL(mission):
  for each capable car: compute ETA after onboard passenger drain
  choose minimum feasible pickup ETA, else mark UNAVAILABLE
  propose exclusive route; Cabin validates and ACKs
ON_APPROVED_FREIGHT(mission):
  select eligible goods or permitted passenger car
  complete onboard destinations; verify empty handoff
  collect approved material; check load and locked doors
  deliver direct to approved floor; restore AUTO after unloading
```

4.8 SATSA-Based Dispatch Optimization (High Priority, Offline)

Description. This feature uses the Simplex-based Adaptive Tree-Seed Algorithm (SATSA) from Liu et al. [8] to search for dispatch grouping configurations, running on the same deterministic elevator simulator as the baseline. The paper presents an optimization method tested on benchmark functions and constrained engineering problems. Its results say nothing about elevator waiting times. So for us, SATSA is a planned, research-grounded feature, and its evaluation and integration are specified for future implementation.

Stimulus/response. Control Center asks for an optimization run, giving an approved set of traffic scenarios and a parameter search domain. The optimization engine generates candidate settings, replays the traffic traces, scores each feasible candidate and returns a versioned recommendation. An authorized reviewer can approve it for the next safe dispatch decision boundary. Routes already accepted keep their existing commitments.

OPT-01 Accept a defined input package containing traffic scenario IDs, fixed simulation seeds, a parameter domain, objective coefficients and the reference Dispatch policy version. Reject incomplete or conflicting optimization requests.

OPT-02 Encode each SATSA candidate as x = (r, w) and project it to the valid integer configuration θ = (R, W) before simulation. Retain the original baseline R = 3 floors and W = 5 s as a reference configuration. Search bounds are proposed, not assumed optimal.

OPT-03 Generate candidate solutions using the three SATSA mechanisms from [8]: adaptive Gaussian seed update, additional seeds for stagnant candidates and Nelder-Mead simplex refinement. Reuse the same random seeds and traffic traces when comparing configurations.

OPT-04 Evaluate candidate configurations by replaying a fixed set of traffic scenarios and computing the defined fitness function. Exclude any run that violates hard route feasibility, Cabin safety validation, protected onboard commitments or cumulative DELAY_CAP.

OPT-05 Compare the chosen SATSA configuration against the fixed (3, 5) baseline and a complete grid search over the same finite (R, W) domain. Report results on separate held-out traces and do not claim improvement without measurement.

OPT-06 Store the selected configuration with its optimization-run ID, input version, objective value and approval status. Only Control Center authorization may activate it, and activation shall occur at a safe configuration boundary without modifying previously accepted route commitments.

4.8.1 Candidate Variables and Feasible Domain

The initial search vector holds only the parameters the real-time algorithm already uses: θ = (R, W). R is the maximum floor span of a same-direction cohort, and W is the fixed time window measured from its anchor call. To keep the optimization contract concrete, we propose a trial domain of R in {1, 2, 3, 4, 5} floors and W in {0, 1, ..., 10} simulated seconds. The team hasn't approved this domain yet, and we aren't presenting it as an experimentally optimized range. SATSA produces continuous search points, so candidates are clamped and rounded to valid integer pairs, and a repeated pair can reuse its cached simulation fitness. This discrete mapping is our adaptation of the continuous method in [8].

4.8.2 Fitness Function and Hard Constraints

For a candidate θ and a fixed set of S simulated traffic scenarios, we compute one score per scenario and average them. Waiting times are in simulation seconds. λ has no units, and μ is in seconds because it multiplies a fraction of unserved requests. λ and μ are fixed and approved before each optimization run, and SATSA doesn't optimize them.

$$J(\theta) = \frac{1}{S}\sum_{s=1}^{S}\Bigg[\ \text{mean_wait}_s(\theta) + \lambda\ \text{P95_wait}_s(\theta) + \mu\,\frac{\text{unserved}_s(\theta)}{\text{total_calls}_s}\Bigg]. \tag{6}$$

$$\theta^* = \arg\min_{\theta\in\Theta_{\text{valid}}} J(\theta), \tag{7}$$

where Θ_{valid} consists of the configurations for which every simulated dispatch step satisfies hard safety, route ownership and DELAY_CAP requirements.

If a candidate breaks a mandatory system invariant, the optimization engine rejects it. It never accepts an unsafe configuration in exchange for a finite penalty. Dispatch still assigns Hall Calls with the existing ETA and directional-insertion policy, and the Cabin still applies its own independent ACK/NACK safety veto. SATSA doesn't change MEDICAL fleet-wide ETA selection or FREIGHT direct transport.

4.8.3 Optimization-to-Deployment Workflow

4.8.4 Readable SATSA Integration Pseudocode

```
OPTIMIZE_DISPATCH_PARAMETERS(training_traces, holdout_traces, bounds, λ,
  μ):
  initialize SATSA trees within bounds; keep reference θ_base = (3, 5)
  repeat until the approved offline evaluation budget is exhausted:
    generate seeds with Gaussian updates and stagnation-triggered expansion
    refine weaker candidate locations with Nelder-Mead simplex steps
    for each continuous candidate x:
      θ = clamp_and_round_to_integer_R_W(x)
      replay the identical training traces with Dispatch configured by θ
      if hard_invariant_violation: reject θ and record why
      else: calculate and cache J(θ); update the best feasible candidate
  compare best θ against θ_base and finite-domain grid-search results
  evaluate all candidates on previously unseen held-out traces
  export versioned recommendation; await Control Center approval
  return θ_star, metrics, experiment_seed, trace_set_version
```

4.9 Worked Examples: Reproducible Calculations

A. One ordinary passenger

P1 is idle at floor 31. A passenger at floor 27 presses DOWN. Pickup ETA = $(31 - 27) \times 2 = 8$ s. After the 6 s pickup dwell, the passenger selects floor 5 and the Cabin travels 22 floors: drop-off arrival = $8 + 6 + (27 - 5) \times 2 = 58$ s. The door dwell at floor 5 finishes at 64 s. The destination isn't used at any point before boarding.

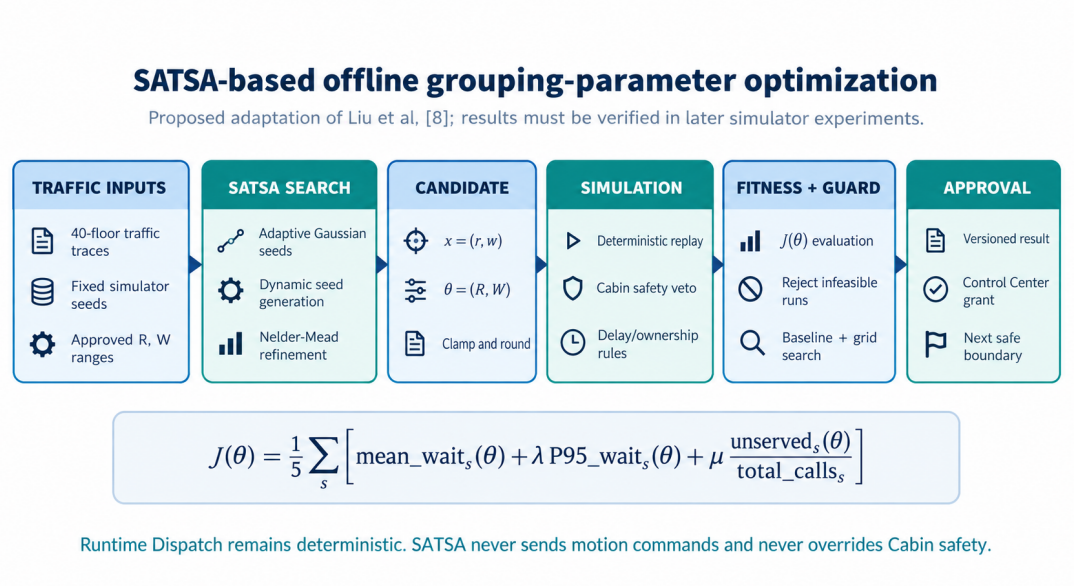


Figure 5: SATSA handles offline parameter selection only. The real-time Dispatch Controller and the local Cabin safety checks keep governing elevator operations.

B. Two passengers, floors 27 and 29 DOWN

Calls arrive at 0 s and 2 s, so the floor span is $2 \le 3$ and the call-time span is $2 \le 5$. An eligible P1 at floor 31 can stop at 29 first (4 s), dwell 6 s and reach 27 at 14 s. Served alone, the older floor-27 request would be picked up at 8 s, so the group adds 6 s to it. That fits within DELAY_CAP = 10 s, though other accepted commitments and Cabin feasibility still have to be checked. Dispatch doesn't hold the assignment waiting for the second call.

C. Medical selection across the whole fleet

Car	Current status	ETA to medical floor 2	Eligible?
P1	Floor 36, onboard destination floor 40	(40−36)×2 + 6 + (40−2)×2 = 90 s	Yes
P2	Idle at floor 12	(12−2)×2 = 20 s	Yes, selected
P3	Floor 5, maintenance	—	No
P4	Floor 28, onboard destination floor 22	(28−22)×2 + 6 + (22−2)×2 = 58 s	Yes
G1	Floor 3, goods-only in this example	—	No

D. Building Staff freight after passenger commitments

P1 starts at floor 27 with a passenger already headed to floor 5. The passenger reaches floor 5 after 44 s and exits during a 6 s stop (t = 50 s). P1 then goes to the material pickup at floor 1 (8 s, t = 58 s). After a 6 s loading dwell (t = 64 s), it runs directly from floor 1 to floor 30 (29 × 2 = 58 s) and arrives at t = 122 s. All of this assumes P1 is the chosen eligible car, no safety event interrupts it and loading is confirmed.

5 Other Nonfunctional Requirements

5.1 Performance Requirements

The numbers below are proposed engineering acceptance targets, not measured results. We still have to show that the integrated simulator meets them under specified load and fault scenarios.

ID	Target / definition
NFR-01	Every accepted Hall Call receives immediate visible acknowledgement; the dispatch loop attempts assignment without deliberate grouping delay.
NFR-02	Every feasible inserted stop preserves all existing reference ETAs within the cumulative 10 s DELAY_CAP.
NFR-03	An AGE_LIMIT of 60 s triggers protection; measure mean, median, P95 and maximum observed pickup waits.
NFR-04	Measure median and worst-case dispatch computation wall time using fixed scenario traces; report unserved and fault-reassigned calls.
NFR-05	The same input event trace, policy version and random seed yield identical route and ownership logs.

ID	Target / definition
NFR-06	Offline SATSA runs shall record candidate configurations, scenario IDs, optimizer seeds, fitness values, invalid candidates, held-out evaluation metrics and Control Center approval decisions.

5.2 Safety Requirements

The Cabin-local veto takes precedence over ordinary work, MEDICAL and FREIGHT. If doors, position validity, motion permission or load constraints fail, unsafe simulated motion is blocked. Nothing in this SRS establishes real-world safety certification.

5.3 Security and Privacy Requirements

Only authorized medical and building roles may submit privileged missions, and a supervisor approves the return from maintenance or a fault hold. Ordinary direction Hall Calls shall not require personally identifying data. The Control Center logs approvals, refusals, emergency alerts and route-version decisions with timestamps.

5.4 Software Quality Attributes

For maintainability, Dispatch, Cabin and Control Center must be separate modules. For testability, we need deterministic event injection and observable state transitions. For reliability, route commands must be idempotent, request IDs stable, and recovery must never produce duplicate assignments.

5.5 Business Rules

Priority order: the safety veto comes before service scheduling; authorized MEDICAL comes before new ordinary or freight work; destinations of passengers already onboard stay committed; and approved freight gets an exclusive direct leg after an empty handoff. Policy can never order an occupied passenger removed from a moving Cabin.

6 Research Basis, Feasibility and Verification

6.1 What Existing Research Supports

Elevator group control is a well-studied scheduling and allocation problem. Siikonen [2] uses models and simulation to analyze dispatch and passenger service in high-rise groups. Cortés et al. [3] look at genetic controller strategies during lunch-peak traffic, and Bolat and Cortés [4] compare genetic and tabu search methods for assigning landing calls to cars. Crites and Barto [5] take a different route, using reinforcement learning agents. These papers show that the problem is worth studying and how to evaluate solutions. They don't show that our 3-floor grouping rule is optimal, or that our implementation is faster than their methods.

6.2 Project Contribution and Design Rationale

Our student implementation uses transparent, reproducible dispatch rules that suit a 40-floor software simulator: filtered candidates, ETA-based choice, direction grouping, a fixed cumulative delay guard, medical selection across the whole eligible fleet, exclusive freight after passengers are dropped, and a Cabin-local safety veto. We chose a deterministic dispatcher because every decision can be explained, replayed and tested without a trained model. We don't claim a new, globally optimal scheduling algorithm.

The small fleet keeps the cost down. An ordered floor index supports O(log F) insertion and lookup for F floors. With C candidate cars and at most K committed stops per car, a simple implementation that rescans every insertion position and recomputes route ETAs takes $O(C \times K^2)$ time per dispatch decision, not counting index operations. That is manageable for a prototype with C = 4 passenger cars, but we still need to benchmark real execution time under generated traffic.

6.2.1 Source Basis and Limits of SATSA Application

Liu et al. [8] introduce SATSA for function optimization and constrained engineering problems. The paper describes adaptive Gaussian seed generation, dynamic updates to seed quantity and location, and Nelder-Mead simplex improvement, and it tests them on CEC2017 and engineering design benchmarks. We adapt SATSA to the finite discrete search domain for the grouping radius R and time window W (Section 4.8). That use is our design proposal, not a result reported in the paper.

The objective, the allowed parameter domain, trace replay, the safety rejection rule and the Control Center approval step are defined in Sections 4.8.1–4.8.4. We left out w_wait and w_age because the real-time Dispatch ranking doesn't use those weights today. SATSA is part of the proposed offline configuration feature and never runs inside the real-time route loop.

Research boundary: The SATSA paper demonstrates an optimization strategy in the domains it tested. We claim no elevator waiting-time improvement until we run scenario experiments.

The design-stage validation plan compares SATSA with the fixed grouping configuration and with exhaustive enumeration over the same bounded two-variable search space. Training traces pick the candidate, and held-out traces are saved for evaluation. The reference calculation script doesn't implement a SATSA optimizer or gRPC transport, and we don't present either as already built.

6.3 Feasibility Evidence: Checked, and Not Yet Checked

Evidence tier	What is currently supported	What is not established
Worked calculations	Exact example ETA, delay and material timeline computed from stated assumptions.	Real elevator travel time or safety compliance.
Executable reference check	Companion Python script asserts four deterministic scenarios; run output reports all four checks passed.	Integrated UI, sensor, RPC or multi-user workload performance.
Literature grounding	Published work supports fleet allocation and simulation as valid research topics [2]–[5].	The superiority of this project's particular algorithm.

To run the reference script: python Smart_Elevator_Dispatch_Feasibility_Demo.py. It checks case A, case B (with cumulative delay and cutoff), case C and case D. This is a calculation-level feasibility check, not a deployed or integrated controller. It doesn't implement SATSA optimization or gRPC transport, so those two design components still need integration evidence later.

6.4 Minimal Acceptance Demonstration

Test	Input / condition	Observable passing result
AT-01	P1 at 31, Hall Call 27 DOWN, Cabin Destination 5	Pickup at 8 s; destination arrival at 58 s; separate lifecycle events
AT-02	Calls 27 and 29 DOWN within 2 s	Group allowed when safe; 27 pickup at 14 s; reject cumulative delay >10 s
AT-03	Medical floor 2; fleet of Figure 3	P2 selected at 20 s; P3/G1 excluded with reason
AT-04	P1 occupied 27→5, freight 1→30	Drop passenger first, pickup material at 58 s, deliver at 122 s
AT-05	Cabin unsafe or route ACK missing	Cabin veto / reconciliation; no duplicate effective owner
AT-06 (design)	Fixed training/hold-out traces; proposed SATSA domain and fixed (3,5) baseline	Every returned (R,W) is in range, violations are rejected, a versioned recommendation is produced, and baseline/grid-search comparison is reportable after implementation

6.5 Evaluation Plan for the Full Prototype

Once everything is integrated, we'll replay identical event traces for three approaches: a first-idle baseline, a directional sweep baseline and our commitment-aware approach. For parameter optimization, we'll compare a fixed R = 3 / W = 5 configuration, an exhaustive search over the approved small discrete domain and SATSA-tuned configurations, all on the same scenarios and held-out seeds. The comparison covers mean and P95 pickup wait, the fraction of unserved calls, onboard delay, medical pickup ETA, freight completion, dispatcher computation time and offline optimization cost. We won't infer performance gains from the SATSA paper, and this SRS doesn't claim any new integrated benchmark result.

6.6 Submission Boundary

This is a design-stage SRS. It includes a research-grounded Dispatch policy, a planned offline SATSA optimization feature, gRPC logical contracts and calculation-level feasibility checks. Sections 7–9 add observable state transitions, an event-driven simulator, backlog requirements and acceptance criteria. None of that is evidence that the integrated system, the SATSA engine or the networked services have been built or benchmarked.

7 Detailed System Models and Simulation Contracts

This section spells out the observable behavior behind the algorithms in Section 4. "Cabin motion", "door operation", "sensor" and "fault" all mean software models and adapters. The physical safety controller, certified brake and rescue procedures sit outside the system boundary. Each rule is written so simulator tests can verify it later, and nothing here implies the system has already passed those tests.

7.1 Behavioral Allocation and System Invariants

The three subsystems keep separate authority. Dispatch owns the canonical request record and proposes routes. Each Cabin Controller validates and runs its accepted route, and supplies the authoritative position, door, load and fault telemetry. Control Center grants privileged missions and watches the fleet, but it can't override a local Cabin safety veto. The Simulation Engine advances time and delivers events to those modules through an ordered event queue.

Domain	Authoritative component	Contract that must remain true
Hall Call ownership	Dispatch Controller	At most one confirmed cabin owns an active (floor, direction) Hall Call.
Cabin route	Cabin Controller	Only the Cabin may ACK a versioned route; the last accepted route remains valid on NACK.
Physical-model state	Cabin / sensor adapters	No movement when door lock, load, position or fault checks fail.
Special mission authorization	Control Center	MEDICAL and FREIGHT require a valid, recorded role authorization.
Simulation clock and event order	Simulation Engine	Equivalent initial states and seeds produce the same ordering of modeled events.
Offline tuning	SATSA Engine + Control Center	Only an approved, versioned configuration reaches a safe future boundary.

SYS-01 The simulator shall model a 40-floor building with passenger cars P1–P4 and goods car G1 as separately identified cabin instances in the baseline configuration.

SYS-02 Each cabin shall maintain independent position, travel direction, motion state, door state, load, operating mode, accepted route and route version.

SYS-03 The simulator shall reject a floor request outside floors 1–40 and shall reject outward-facing terminal Hall Calls (DOWN at floor 1 and UP at floor 40).

SYS-04 The application shall use simulated actuator and sensor adapters; no module shall send commands to installed lift machinery in this project.

7.2 Hall Call, Cabin Destination and Mission Lifecycles

An ordinary Hall Call identifies a floor and the direction the person wants to go. It isn't a passenger itinerary, because the destination isn't known until the passenger boards and picks a Cabin Destination. A privileged mission is different: an authorized request can give pickup and delivery floors before assignment. Mixing the three records would produce false waiting times and duplicate completions.

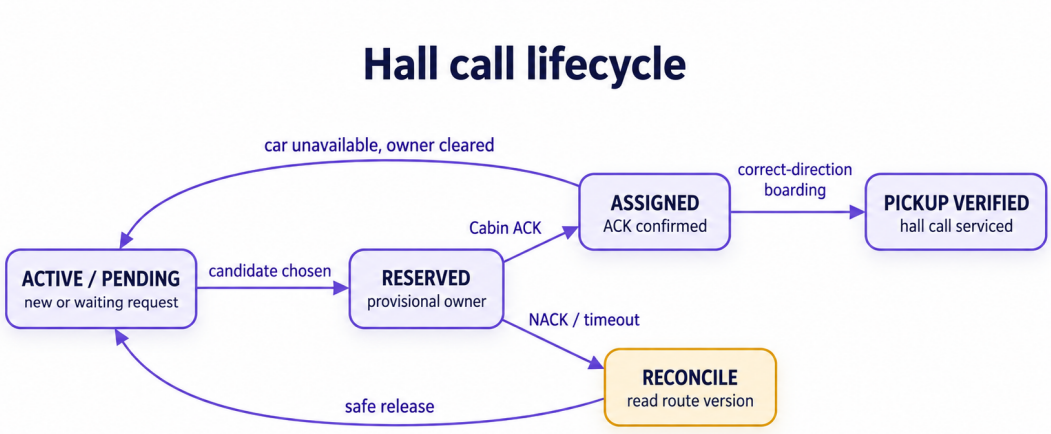


Figure 6: Canonical Hall Call lifecycle with provisional reservation and safe ACK/NACK reconciliation.

Record	Creation	Completion / retention condition
HallCall	Valid UP/DOWN press; deduplicated per floor-direction pair	Service opportunity verified at origin and requested direction; never wait for destination drop-off.
CabinDestination	Passenger boards and selects an allowed destination	Cabin arrives, opens its doors at that floor and the simulated passenger exits.
PrivilegedMission	Authorized medical/staff workflow with known pickup and delivery	Delivery/unloading confirmed or mission explicitly canceled/failed under approved policy.

REQ-01 Every active Hall Call shall have request_id, origin_floor, direction, created_at, status and confirmed_owner (nullable).

REQ-02 A duplicate Hall Call at the same active floor and direction shall return the existing request_id without creating an independent owner or resetting created_at.

REQ-03 A route proposal shall be associated with a provisional reservation; a confirmed assignment shall be recorded only after the addressed Cabin ACKs the accepted route version.

REQ-04 Following NACK or timeout, Dispatch shall reconcile the last accepted route and provisional ownership before another Cabin is offered the same call.

REQ-05 The simulator shall record Hall Call pickup separately from the Cabin Destination arrival and unloading event.

Example. Two people press DOWN at floor 27 within the same minute. The physical hall panel shows one active 27/DOWN call. The simulator may track two virtual passenger entities for boarding statistics, but it shall not erase the active hall call just because one person leaves the queue. Without an identity-aware passenger interface, canceling an individual Hall Call is not a core requirement.

7.3 Cabin Motion and Direction Model

A Cabin Controller works from a route that has already passed Dispatch's feasibility screen and its own local checks. The motion model tracks continuous position p(t), even though the passenger interface shows discrete floor labels. A floor label is only a presentation value. It isn't a reliable measurement of braking or stopping position.

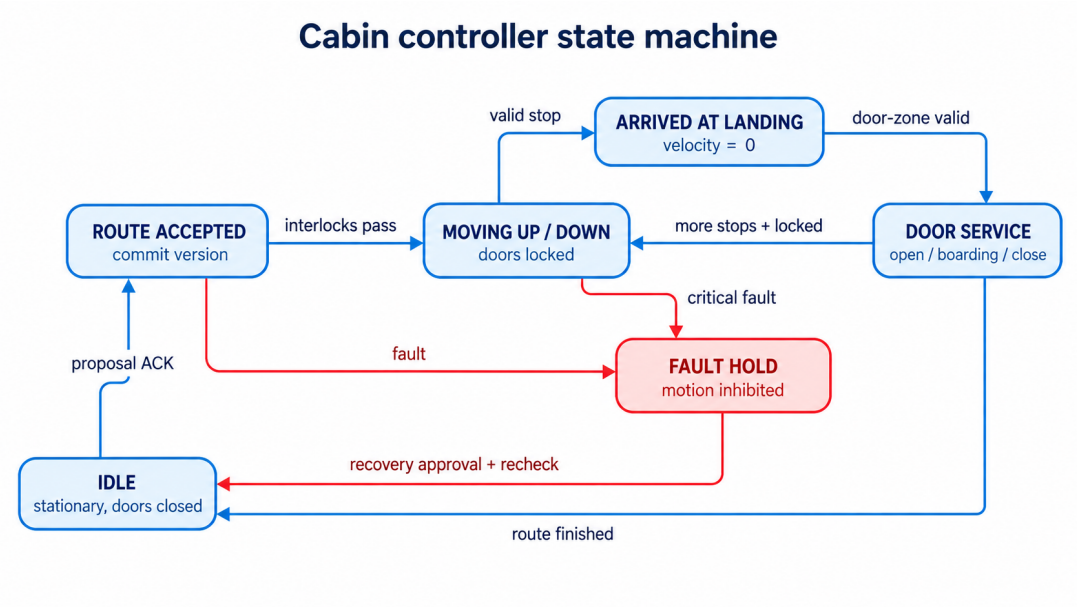


Figure 7: Cabin operating-state model; a serious modeled fault inhibits movement until authorized recovery.

MOVE-01 The Cabin shall report continuous simulated position, signed direction (UP, DOWN or IDLE), velocity state and timestamp with each relevant telemetry update.

MOVE-02 The Cabin shall initiate modeled movement only when a route is accepted, its doors are locked, load and position readings are valid and no inhibiting fault exists.

MOVE-03 The Cabin shall not teleport from one floor to another during normal simulation; it shall advance position with the configured motion model and generate an arrival event at a valid served floor.

MOVE-04 When traveling UP, eligible UP Hall Calls ahead may be inserted if the stop is safely reachable and every protected ETA remains within DELAY_CAP; analogous rules apply to DOWN travel.

MOVE-05 The Cabin shall not reverse direction while movement remains committed in the current direction unless a modeled fault or explicitly permitted recovery rule requires a safe hold.

MOVE-06 A moving Cabin shall decline a newly inserted stop when the remaining modeled distance in its current travel direction is less than the configured stopping cutoff or the proposal violates another local constraint.

The current demonstration uses a conservative one-floor insertion cutoff as a simulation-only simplification. Say a car at p = 26.8 is traveling UP. Floor 27 is only 0.2 floors ahead, so a late insertion is rejected even if the rounded display reads floor 26. Real-world braking distance depends on speed, acceleration and hardware characteristics, and this student software doesn't certify any of that.

Condition	Required cabin behavior	Observable event
Valid intermediate stop ahead	Open a candidate insertion check; accept only if route and ETA guards pass	ROUTE_ACK or ROUTE_NACK(reason)
Opposite-direction call while moving	Leave it pending or allocate another eligible car; no instant reversal	CALL_PENDING / OTHER_CAR_ASSIGNED

Condition	Required cabin behavior	Observable event
At a valid committed stop	Stop, verify landing zone and initiate door opening	CAR_ARRIVED, DOOR_OPENING
No outstanding route and no exclusive mission	Set direction to IDLE and report current location	CAR_IDLE

7.4 Door State Machine and Passenger Interaction

Door logic is a local Cabin concern. Dispatch can schedule a stop, but it never opens a physical or simulated door directly. When an occupant asks for the door to open, the request faces the same stopped-at-landing and safe-door-zone conditions as an automatic one.

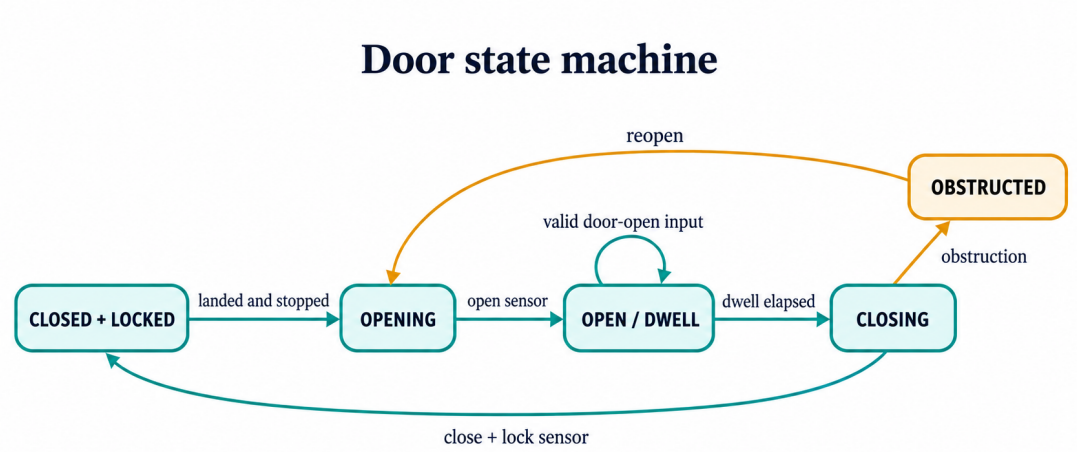


Figure 8: Door control states and obstruction handling; movement permission is available only after the CLOSED + LOCKED state is verified.

DOOR-01 The door model shall expose CLOSED, OPENING, OPEN, CLOSING and OBSTRUCTED states as independent state values.

DOOR-02 The Cabin shall open simulated doors at an approved stop only when it is stationary and the door-zone sensor identifies the served landing.

DOOR-03 The system shall keep an OPEN door state for a configurable simulated dwell interval unless a permitted request extends that interval.

DOOR-04 When an obstruction occurs during closing, the door model shall stop its closing transition and reenter OPENING; the Cabin shall not enable travel until closed-and-locked confirmation returns.

DOOR-05 The Cabin shall reject a normal door-open command during movement, between floors or during a motion-inhibiting fault that requires doors locked.

DOOR-06 A cabin panel destination pressed before valid passenger boarding shall not be interpreted as a destination for an unknown hall caller; passenger destinations become active only through the cabin interaction model.

Typical transition. The Cabin arrives at floor 27, confirms it is stationary and in the door zone, and issues OPENING. Once the OPEN sensor reports in, the dwell timer starts. When it times out, the door goes to CLOSING and has to see the closed-and-locked sensor before MOVING can resume. If an obstruction flag appears during CLOSING, the timer and state follow the re-open branch. The event log has to make this progression easy to inspect.

7.5 Sensor, Capacity and Empty-Cabin Handoff

The simulator treats the sensors below as event sources. Invalid readings must block or defer an operation. They must never be silently turned into a safe value. MEDICAL and FREIGHT capability checks are configuration rules, separate from sensor integrity.

Simulated input	Used for	Invalid or conflicting reading
Position and door-zone	Travel progress, permitted stops and safe opening	No new stop acceptance / hold the cabin as appropriate
Door closed + locked	Permission to move	Movement blocked
Door obstruction	Safe reversal during closing	Re-open and generate alert event

Simulated input	Used for	Invalid or conflicting reading
Load / overload	Capacity and service eligibility	Block departure above MAX_LOAD
Onboard passenger count + exit events	Confirming drain for privileged missions	Keep the mission in DRAINING until reconciled
Fault/heartbeat freshness	Eligible fleet selection	Exclude stale or faulted cabins

SEN-01 The simulator shall publish position, door-zone, lock, load, obstruction and fault readings with timestamps associated with a unique car_id.

SEN-02 A reading identified as missing, stale or mutually inconsistent shall not be treated as proof of an unlocked condition being resolved.

SEN-03 The Cabin shall refuse to initiate motion when modeled load exceeds MAX_LOAD = 1000 kg; after reaching FULL_LOAD = 900 kg it shall not accept new ordinary pickups until load permits.

SEN-04 When overloaded, the cabin shall remain stationary; a virtual passenger may explicitly exit at the current landing, and any later request for that passenger shall originate at the actual exit floor.

SEN-05 An exclusive MEDICAL or FREIGHT handoff shall require the Cabin to be stationary at a served landing, onboard passenger count equal to zero, completed exit events and non-conflicting load/door-zone evidence.

SEN-06 Conflicting passenger-count and load observations shall keep the mission in DRAINING or HOLD_PENDING_REVIEW and notify Control Center; the software shall not assume the cabin is empty solely because a thresholded load sensor reads low.

7.6 Safety Precedence, Emergency and Recovery

The system treats an authorized MEDICAL dispatch mission, an occupant's alarm and a fire or motion-safety event as three different things. MEDICAL is a scheduling priority among feasible service jobs. A cabin alarm or safety hold is an exceptional equipment or occupant event. Keeping them apart stops a routine alarm from being read as a medical route grant.

Event type	Dispatch response	Cabin response	Control Center response
MEDICAL mission approved	Select fleet-wide earliest feasible pickup	Complete onboard commitments; accept exclusive mission only after empty handoff	Authorize and monitor mission
Passenger ALARM	Retain current ownership; raise alert	Follow configured alarm/safe-stop policy; do not self-grant MEDICAL	Acknowledge alarm and coordinate operator action
Serious fault between floors	Reconcile and reassign unpicked external calls	Inhibit simulated motion; keep doors locked between floors	Notify operator; require authorized recovery
Fire-recall input	Suspend ordinary planning under scenario policy	Apply configured simulated fire response subject to local veto	Display fire alert and recovery state
Overload at landing	Exclude new pickup while full	Remain at landing until safely resolved	Display load warning

EMG-01 The system shall expose separate ALARM, MEDICAL_MISSION, FIRE_ALERT and FAULT_HOLD events and shall not automatically map an ordinary ALARM to privileged medical dispatch.

EMG-02 On a serious modeled between-floor fault, Cabin shall enter FAULT_HOLD, inhibit movement and normal door opening, retain onboard passenger records and alert the operator.

EMG-03 Dispatch shall preserve the creation time of each unpicked Hall Call reassigned after a cabin fault; the faulted cabin shall not be reconsidered until recovery approval and successful health checks.

EMG-04 A supervisor shall authorize return from FAULT_HOLD or MAINTENANCE to AUTO only after the simulation indicates all specified readiness checks have passed.

EMG-05 A MEDICAL mission that remains unserved shall show a persistent operator-visible state; any automatic expiry policy requires an explicit, approved escalation or cancellation rule and shall not silently drop the emergency task.

7.7 Deterministic Simulation Engine and Event Ordering

An evaluator should be able to replay a passenger scenario and explain each assignment, without assuming the network is perfectly synchronous. The Simulation Engine supplies a discrete-event clock. When events share the same simulated timestamp, it uses a stable order with an ever-increasing sequence number. Decisions use immutable event snapshots, not arbitrary UI animation frames.

Event category	Examples	Scheduling rule
Safety input	fault, obstruction, overload, door lock loss	Deliver before a same-timestamp movement or route commitment
Cabin/physical-model result	arrival, door opened, exit event, telemetry	Update authoritative cabin state before selecting new route

Event category	Examples	Scheduling rule
Control message	route proposal, ACK/NACK, mission approval	Process by (time, priority, sequence) with idempotent command ID
Passenger input	Hall Call, Cabin Destination	Validate and store as a single canonical event
Reporting	metrics, dashboard refresh, log export	Observe post-transition snapshot; do not alter decisions

SIM-01 The simulator shall maintain a controllable simulation clock and shall expose RUNNING, PAUSED and STOPPED session states.

SIM-02 While PAUSED, no simulated clock tick, movement progress, door transition or scheduled event execution shall advance; current state shall remain visible.

SIM-03 The operator shall be able to reset to a declared initial fleet configuration and replay a saved event trace and seed.

SIM-04 Events shall be ordered deterministically by simulated time, fixed event-priority category and insertion sequence number.

SIM-05 The simulator shall accept new Hall Calls while all passenger cabins are moving and shall allow the scheduler to report PENDING when no immediately feasible route exists.

SIM-06 Every RouteProposal shall carry command_id and expected_route_version so a repeated delivery can return the stored ACK/NACK result without a second route execution.

SIM-07 On missing ACK, Dispatch shall not grant a second confirmed owner until reconciliation determines whether the first proposed route was accepted.

SIM-08 Simulated hardware adapters shall publish feedback events rather than directly rewriting Dispatch request ownership.

Example ordering. At t = 20 s, an obstruction input and a door-close completion are both due. The safety input is processed first, so the completed-close event may be invalidated and the door heads back toward OPEN. In a separate case, a RouteProposal times out while its original ACK is delayed. Dispatch then asks for the authoritative route version before reassigning anything. These are the outcomes we expect from the design, and they must be verified during integration.

7.8 User Interfaces and Monitoring Information

The eventual frontend team may draw the UI differently, but what it shows is part of the requirements. Anyone should be able to understand fleet status without reading low-level log messages, and every privileged action must show whether it is requested, approved, awaiting handoff, in progress or completed.

Surface	Minimum controls	Visible feedback
Hall Panel	UP / DOWN, permitted by floor	Request accepted; waiting / assigned indication
Cabin Panel	Floor buttons 1–40, door requests, alarm	Current floor, destination selection, direction, door and overload state
Control Center	Medical and freight approval, fleet view, fault recovery	Car mode/position/load, mission state, unavailable conditions and audit record
Simulation Console	Start, pause, reset, replay, fixed seed selection	Clock time, pending events, configuration version, basic metrics
Optimization View	Scenario set and SATSA run request; approval action	Candidate θ, fitness, comparison against (3,5), approved version

UI-01 The main fleet view shall display P1–P4 and G1 independently with current position, direction, door status, load warning, mode and active route.

UI-02 The hall-panel display shall distinguish an accepted pending call from an assigned call and shall remove its active indicator only after verified service.

UI-03 A passenger shall see valid destination selections and each selected Cabin Destination shall have distinct pending/served visual states.

UI-04 The Control Center shall display an explicit NO_ELIGIBLE_CAR or UNAVAILABLE status instead of presenting a failed privileged assignment as successful.

UI-05 The simulation console shall show session state and simulation time and shall permit the tester to inspect request ownership and accepted route versions.

7.9 Event Logging, Metrics and Replay Evidence

We want a compact, machine-readable event log, because traceability depends on seeing both the decisions and the safety vetoes. The core fields are event_id, simulated_time, event_type, actor, car_id where relevant, request_id or mission_id, previous_state, new_state, route_version and an optional reason. Sensitive mission attributes shall be role-limited. Ordinary Hall Calls need no personal identifier.

00:00:00 HALL_CALL_ACCEPTED call=HC27D origin=27 dir=DOWN

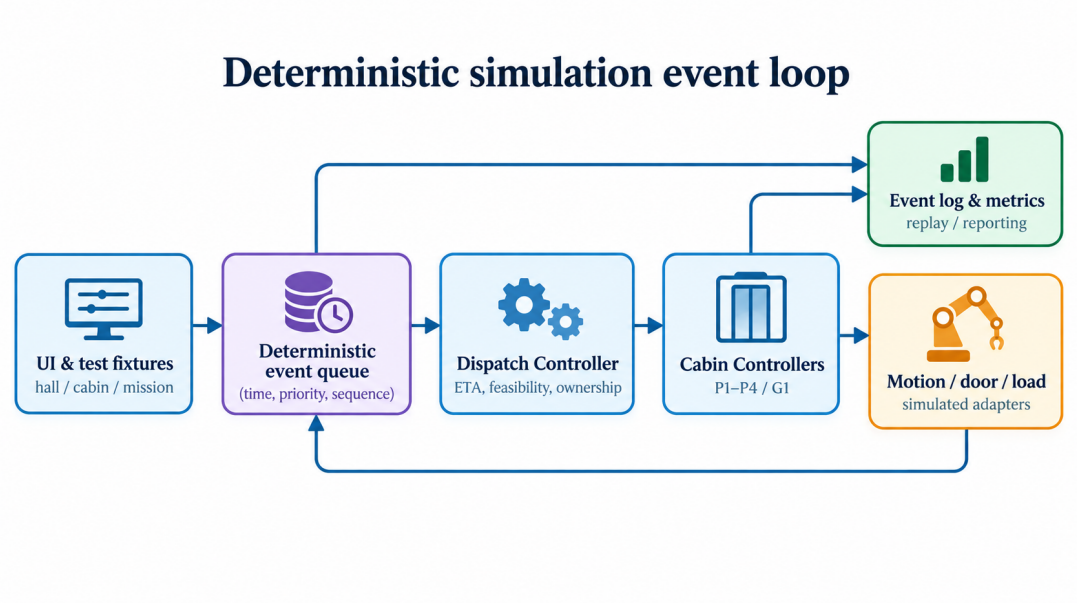


Figure 9: Event-driven simulation components and the feedback path from cabin sensor adapters to the next dispatch decision.

00:00:00 ROUTE_PROPOSED car=P1 cmd=RC004 expected_version=7
00:00:00 ROUTE_ACK car=P1 cmd=RC004 committed_version=8
00:00:08 CABIN_ARRIVED car=P1 floor=27
00:00:08 HALL_CALL_SERVED call=HC27D car=P1
00:00:14 CABIN_DESTINATION_ACCEPTED car=P1 floor=5

LOG-01 The system shall log each Hall Call acceptance/deduplication, proposed route, ACK/NACK, verified pickup, Cabin Destination completion, mission transition and fault state transition.

LOG-02 A stored trace and configuration version shall be sufficient to reproduce simulated order-dependent decisions under the same model revision and random seed.

LOG-03 The simulator shall record at least mean and P95 Hall Call pickup wait, unserved-call count, commitment delay violations, medical pickup time and freight mission completion time for later evaluation.

LOG-04 The system shall record SATSA optimization runs separately from active Cabin route execution, including objective parameters, input trace set, candidate θ and approval status.

8 Agile Product Backlog and Release Plan

The 44-page Module 5 teaching handout [10] describes a hybrid approach: one system-level SRS supported by Agile epics, user stories, acceptance criteria and traceability. The backlog below follows that format and doesn't replace the normative requirements in Sections 4 and 7. "Given/When/Then" statements describe observable behavior we expect in the future. They are not PASS results, and they don't mean the software exists yet.

8.1 Backlog Conventions and Prioritization

Priority	Meaning	Examples
P0 – Safety / contract	Without this, a credible multi-cabin demonstration is not possible	Door interlock, one owner per call, route ACK, fault hold
P1 – Core operation	Required for the baseline project objective	Normal dispatch, directional grouping, cabin destination, monitoring
P1 – Research feature	Explicit core of the group's intended project design	SATSA offline tuner and controlled configuration approval
P2 – Presentation refinements	Useful after the behavior contracts are fixed	Additional dashboards, report exports and extended scenario catalogs

8.2 E1 – Building and Cabin Modeling (P0)

8.2.1 US-001 Represent the Building

As a simulation operator, I want a 40-floor building and separately identified P1–P4/G1 cars so that I can inspect fleet behavior on a consistent baseline.

AC-001-1: Given a baseline session starts, when the system loads, then floors 1–40 and cars P1–P4/G1 are present.

AC-001-2: Given any car changes position, when telemetry is processed, then the displayed position and mode update independently of other cars.

Related requirements: SYS-01, SYS-02, UI-01.

8.2.2 US-002 Control Simulation Time

As a tester, I want to pause, reset and replay a fixed input trace so that I can check the same design decision under repeatable conditions.

AC-002-1: Given a running session, when PAUSE is requested, then simulated motion and event execution stop.

AC-002-2: Given a saved seed and input trace, when replay starts from the same initial state, then the recorded event sequence is repeatable.

Related requirements: SIM-01, SIM-02, SIM-03, LOG-02.

8.3 E2 – Ordinary Passenger Requests (P1)

8.3.1 US-003 Create a Hall Call

As a passenger, I want to press UP or DOWN on the hall panel so that the system calls a suitable cabin without needing my destination in advance.

AC-003-1: Given floor 27, when DOWN is pressed, then one canonical 27/DOWN request appears.

AC-003-2: Given 27/DOWN is already active, when DOWN is pressed again, then the same request ID is returned and its original timestamp is retained.

AC-003-3: Given floor 1, when DOWN is pressed, then the terminal-direction input is rejected.

Related requirements: DC-01, REQ-01, REQ-02, SYS-03, UI-02.

8.3.2 US-004 Board and Select a Destination

As a passenger, I want to select my destination after boarding so that the cabin can schedule my trip without the dispatcher guessing it.

AC-004-1: Given a verified pickup at floor 27, when floor 5 is selected, then a Cabin Destination(5) is created.

AC-004-2: Given the Cabin reaches floor 5 and the simulated passenger exits, then the destination is complete while the Hall Call remains separately recorded as served.

Related requirements: CB-03, CB-04, REQ-05, DOOR-02.

8.4 E3 – Fleet Dispatch and Route Ownership (P0/P1)

8.4.1 US-005 Choose a Feasible Cabin

As a passenger, I want the system to evaluate all eligible passenger cars so that I receive a realistic assignment that respects accepted routes.

AC-005-1: Given a new Hall Call, when Dispatch evaluates cabins, then unavailable or incompatible cabins are excluded.

AC-005-2: Given multiple feasible routes, when candidates are ranked, then the earliest feasible pickup ETA wins subject to the defined tie break.

AC-005-3: Given an existing accepted route, when a candidate would exceed the 10 s cumulative delay cap, then that insertion is rejected.

Related requirements: DC-02, DC-04, DC-05, MOVE-04.

8.4.2 US-006 Group Same-Direction Requests

As a passenger, I want a compatible passing cabin to collect nearby same-direction calls so that intermediate pickup opportunities can be shared without a deliberate hold.

AC-006-1: Given requests 27/DOWN at t=0 and 29/DOWN at t=2, when route feasibility passes, then they may belong to the same three-floor/five-second cohort.

AC-006-2: Given a request at t=0, when no second request exists, then Dispatch shall still attempt immediate assignment.

Related requirements: DC-03, DC-04, MOVE-04.

8.4.3 US-007 Prevent Duplicate Ownership

As a building operator, I want each Hall Call to have one confirmed owner so that two cabins do not claim an exclusive pickup independently.

AC-007-1: Given one call and a RouteProposal without ACK, when a timeout occurs, then the previous route is reconciled before another owner is confirmed.

AC-007-2: Given the same command_id is replayed, when Cabin receives it, then it returns the stored result without committing a second route.

Related requirements: SAF-04, REQ-03, REQ-04, SIM-06, SIM-07.

8.5 E4 – Cabin Safety, Doors and Faults (P0)

8.5.1 US-008 Move With Door Interlocks

As a passenger, I want the cabin to move only after the doors lock so that the simulated trip does not depart in an unsafe state.

AC-008-1: Given the doors are OPEN, when movement is proposed, then the Cabin rejects departure.

AC-008-2: Given a valid stop at a landing, when the Cabin is stationary in the door zone, then opening may begin.

Related requirements: SAF-01, MOVE-02, DOOR-02, DOOR-05.

8.5.2 US-009 Handle an Obstructed Door

As a passenger, I want a closing door to reopen on obstruction so that the simulation handles the expected door safety event.

AC-009-1: Given CLOSING, when obstruction becomes active, then the door returns to OPENING.

AC-009-2: Given any state other than CLOSED + LOCKED, when a movement command arrives, then movement remains inhibited.

Related requirements: DOOR-01, DOOR-04, SEN-01.

8.5.3 US-010 Recover From a Cabin Fault

As a Control Center supervisor, I want a failed cabin to be isolated while others keep serving so that one local failure does not stop the entire group.

AC-010-1: Given P1 faults between floors, when telemetry is received, then P1 enters FAULT_HOLD and cannot accept normal work.

AC-010-2: Given P2 is healthy, when P1's unpicked calls are reconciled, then they may be reassigned without deleting their original timestamps.

AC-010-3: Given recovery approval and passing health checks, then P1 may return to AUTO.

Related requirements: SAF-03, EMG-02, EMG-03, EMG-04.

8.6 E5 – Privileged Missions (P1)

8.6.1 US-011 Dispatch an Authorized Medical Mission

As a medical coordinator, I want a permitted medical pickup to be prioritized across the fleet so that a compatible cabin arrives as early as feasible without violating passenger commitments.

AC-011-1: Given approved MEDICAL at floor 2 and candidate ETAs P1=90, P2=20, P4=58 seconds, when all checks pass, then P2 is selected.

AC-011-2: Given no eligible cabin, when selection runs, then the mission remains visible as UNAVAILABLE and an alert is raised.

AC-011-3: Given medical boarding is verified, then the reserved medical leg excludes ordinary intermediate pickups.

Related requirements: MED-01, MED-02, MED-04, MED-05, EMG-05.

8.6.2 US-012 Transport Building Materials

As an authorized Building Staff member, I want materials moved between approved floors so that the selected cabin finishes its passenger commitments first and then makes an exclusive delivery.

AC-012-1: Given P1 carries a passenger to floor 5, when the staff mission is approved, then P1 finishes that onboard trip before handoff.

AC-012-2: Given the cabin is confirmed empty at floor 1 and loading checks pass, when the delivery leg begins, then the cabin goes directly to floor 30 without ordinary pickups.

AC-012-3: Given unloading is confirmed, then the mission completes and the exclusive reservation is released.

Related requirements: FRT-01, FRT-02, FRT-03, FRT-04, SEN-05.

8.7 E6 – Communication, Monitoring and Evidence (P0/P1)

8.7.1 US-013 Review Fleet and Mission Status

As a Control Center operator, I want the modes, positions, faults and mission states on one screen so that I can understand what the system is doing and identify unavailable service.

AC-013-1: Given a cabin enters FAULT_HOLD, when telemetry arrives, then Control Center shows the affected car and an alert.

AC-013-2: Given a privileged mission has no eligible car, then its status remains UNAVAILABLE rather than showing assignment success.

Related requirements: CC-02, UI-01, UI-04, LOG-01.

8.7.2 US-014 Exchange Versioned Route Commands

As a backend integrator, I want a typed route proposal and ACK/NACK protocol so that Cabin route decisions are repeatable across service boundaries.

AC-014-1: Given a RouteProposal with an outdated expected version, when Cabin validates it, then Cabin returns NACK without replacing its route.

AC-014-2: Given an identical retried command_id, when it is received again, then the earlier outcome is returned and no second route is committed.

Related requirements: CB-01, REQ-03, SIM-06, SIM-07.

8.8 E7 – SATSA Research Optimization (P1)

8.8.1 US-015 Tune Grouping Parameters Offline

As a research team member, I want SATSA to evaluate finite grouping candidates against fixed traffic traces so that the group can study parameter choices with a reproducible objective.

AC-015-1: Given approved scenarios and seeds, when SATSA proposes real values (r,w), then the simulator projects them to a valid integer (R,W) configuration.

AC-015-2: Given any candidate violates a hard safety or commitment constraint, when fitness is computed, then that candidate is rejected.

AC-015-3: Given a recommendation is produced, then results are compared with baseline (3,5) and exhaustive grid search on held-out traces before a performance claim.

Related requirements: OPT-01, OPT-02, OPT-03, OPT-04, OPT-05.

8.8.2 US-016 Approve Optimized Configuration

As a Control Center supervisor, I want a versioned recommended configuration to require approval so that ongoing routes remain stable when policy parameters change.

AC-016-1: Given a candidate configuration is unapproved, when Dispatch runs, then its active policy is unchanged.

AC-016-2: Given a recommendation is approved, when the next safe dispatch boundary occurs, then the new version becomes available for future decisions without modifying accepted routes.

Related requirements: OPT-06, CC-01.

8.9 Initial Agile Release Plan

Increment	Functional focus	Stories	Observable milestone
R1 – Fleet foundations	Models, simulation clock, Hall Calls and minimal UI	US-001, US-002, US-003	40 floors and five identified cabins; deterministic event replay
R2 – Normal service	Route ACK, ETA, cabin destination, door interlocks	US-004, US-005, US-007, US-008, US-009, US-014	One passenger can request, board and arrive without duplicate ownership
R3 – Grouping and resilience	Same-direction insertion, fault/overload behaviors	US-006, US-010, US-013	Multi-car service persists under fault injection and valid group calls
R4 – Authorized missions	Medical and Building Staff workflows	US-011, US-012	Two privileged workflows satisfy empty-handoff and exclusive-leg rules
R5 – Offline optimization	SATSA search, approval, report	US-015, US-016	Parameter proposal, safe activation and measured comparison design

These release milestones show the order we plan to build things in. They don't mean any increment has shipped. Because this is a design-only submission, the acceptance criteria are what a later implementation has to demonstrate.

9 Requirements Traceability and Acceptance

Traceability links this research-grounded design to the future implementation. Every new SYS/REQ/MOVE/DOOR/SEN/EMG/SIM/UI/LOG requirement is assigned to a subsystem or simulation component. The matrix below samples the main behavior contracts from Sections 4 and 7, and links each to a user story and a verification method. Test labels name planned checks, and execution status stays NOT RUN unless it's independently measured.

9.1 Requirement-to-Story Traceability Matrix

Requirement ID	Owner	Story	Acceptance criteria	Proposed test
SYS-01 / SYS-02	Fleet model	US-001	AC-001-1/2	TC-MODEL-01
SIM-01 / SIM-02	Simulation Engine	US-002	AC-002-1	TC-SIM-01
DC-01 / REQ-01	Dispatch	US-003	AC-003-1/3	TC-HALL-01
REQ-02	Dispatch	US-003	AC-003-2	TC-HALL-02
CB-03 / CB-04 / REQ-05	Cabin	US-004	AC-004-1/2	TC-CAB-01
DC-02 / DC-05	Dispatch	US-005	AC-005-1/2	TC-ETA-01
DC-04	Dispatch	US-005	AC-005-3	TC-DELAY-01

Requirement ID	Owner	Story	Acceptance criteria	Proposed test
DC-03	Dispatch	US-006	AC-006-1/2	TC-GROUP-01
SAF-04 / REQ-03/04	Dispatch + Cabin	US-007	AC-007-1	TC-OWN-01
SIM-06 / SIM-07	Dispatch + Cabin	US-007/014	AC-014-2	TC-IDEMP-01
MOVE-02 / SAF-01	Cabin	US-008	AC-008-1	TC-MOVE-01
DOOR-02 / DOOR-05	Cabin	US-008	AC-008-2	TC-DOOR-01
DOOR-04	Cabin	US-009	AC-009-1	TC-DOOR-02
SAF-03 / EMG-02/03	Cabin + Dispatch	US-010	AC-010-1/2	TC-FAULT-01
CC-04 / EMG-04	Control Center	US-010	AC-010-3	TC-RECOVERY-01
MED-01 / MED-02	Dispatch	US-011	AC-011-1	TC-MED-01
MED-04 / EMG-05	Control Center	US-011	AC-011-2	TC-MED-02
MED-05	Cabin	US-011	AC-011-3	TC-MED-03
FRT-01 / FRT-02	Control + Cabin	US-012	AC-012-1	TC-FRT-01
FRT-03 / FRT-04	Cabin	US-012	AC-012-2/3	TC-FRT-02
UI-01 / UI-04 / CC-02	Control Center	US-013	AC-013-1/2	TC-UI-01
CB-01 / SIM-06	Cabin	US-014	AC-014-1/2	TC-RPC-01
OPT-01 to OPT-04	SATSA Engine	US-015	AC-015-1/2	TC-OPT-01
OPT-05	Research Simulator	US-015	AC-015-3	TC-OPT-02
OPT-06	Control Center	US-016	AC-016-1/2	TC-OPT-03
LOG-01 / LOG-03	Log + Metrics	US-013	AC-013-1	TC-LOG-01

The matrix is a selected coverage view. It doesn't claim that 26 checks verify the whole application. During development, each normative requirement should have exactly one stable identifier, a recorded owner, at least one acceptance method and an execution/evidence status. This design-stage submission deliberately has no test results.

9.2 Representative Acceptance Scenarios

9.2.1 TC-HALL-01 / TC-GROUP-01: Ordinary shared-direction service

Initial state: four passenger cabins are available according to the scenario; P1 is at floor 31, and a passenger requests 27/DOWN at t=0. Another requests 29/DOWN at t=2. Input: two Hall Calls with origins and directions only. Expected: both calls are validated and no destination is assumed. P1 may serve floor 29 and then 27 if route feasibility and cumulative delay allow it. The original t=0 call isn't held for five seconds just to form a group.

Calculation oracle: the span is $|29-27|=2$ floors and the age span is 2 s, so the grouping eligibility values satisfy R=3 and W=5. P1 can reach floor 29 after 4 s, dwell 6 s, and arrive at floor 27 after 14 s. A direct arrival at floor 27 would be 8 s. The extra 6 s must be checked against all accepted reference ETAs and the fixed 10 s cap. Meeting the grouping condition alone doesn't prove the route can actually run.

9.2.2 TC-MED-01: Fleet-wide medical ETA selection

Initial state: a MEDICAL pickup at floor 2 is authorized. P1 is at floor 36 with an onboard destination of floor 40. P2 is idle at floor 12. P3 is under maintenance at floor 5. P4 is at floor 28 with an onboard destination of floor 22. G1 is goods-only in this scenario. Expected: P1=90 s, P2=20 s, P4=58 s. P3 and G1 are excluded. P2 is recommended, and it's assigned only after its Cabin ACK. If P2 NACKs, the route version has to be reconciled before P4 or P1 is considered.

9.2.3 TC-FRT-02: Committed passenger then direct freight leg

Initial state: P1 carries a passenger from floor 27 to floor 5, and staff submit an approved material mission from floor 1 to floor 30. Expected: the passenger is dropped at floor 5. The cabin then goes to floor 1, confirms an empty handoff, loads and closes safely, and delivers to floor 30 with no ordinary stops. At 2 s/floor and 6 s per served stop, the simplified case works out to passenger arrival at t=44, drop-off complete at t=50, pickup arrival at t=58, loading complete at t=64 and freight arrival at t=122. This is a model arithmetic oracle, not a measured time.

9.2.4 TC-DELAY-01: Cumulative commitment protection

Initial state: a passenger destination has a committed ETA of 20 s. Insertion A produces an ETA of 26 s, a +6 s delay, and may be accepted if every other constraint passes. Insertion B would produce an ETA of 32 s. The right comparison is $32-20=12$ s against the original reference, not $32-26=6$ s against the last reroute. Expected: B is rejected because 12 exceeds DELAY_CAP=10 s.

9.2.5 TC-OWN-01: Lost ACK and retry

Initial state: Dispatch has reserved Hall Call HC27D for a candidate Cabin and sends command RC004 at expected route version 7. The Cabin may have accepted the route and moved to version 8, but the ACK packet is delayed. Expected: Dispatch doesn't assign HC27D to another cabin until it has reconciled the actual route version. If RC004 is sent again, it returns the original result and doesn't duplicate the accepted route.

9.2.6 TC-OPT-01: SATSA study contract

Initial state: an operator authorizes an offline tuning run over an approved discrete (R,W) domain, with fixed training seeds and objective coefficients. Expected: SATSA generates candidate points and projects them to valid integers, simulations

compute J(θ), unsafe or infeasible settings are rejected, and the result is compared with both the fixed (3,5) setting and full-grid enumeration. Control Center may approve only a versioned result for future decisions, and no existing Cabin route is rewritten. A measured improvement is something we hope future trials show. This scenario doesn't assume one.

9.3 Boundary and Edge-Case Coverage

Edge case	Expected design response	Requirement link
Repeated hall-panel input	Return existing canonical call without resetting its age	REQ-02
Opposite-direction call on a moving cabin	Decline unsafe insertion or evaluate another cabin	MOVE-04/05
New call when all cabins busy	Keep active as PENDING and reevaluate on state change	SIM-05, DC-06
Door obstruction during closing	Reopen; deny motion until lock confirmation	DOOR-04
Overload with manual exit	Block movement; requeue virtual passenger at actual exit landing	CB-05, SEN-04
Low load reading but nonzero passenger count	Keep mission handoff blocked and request review	SEN-05/06
Critical fault between floors	Cabin FAULT_HOLD; reassign only unpicked external calls	EMG-02/03
No compatible MEDICAL car	Display UNAVAILABLE and alert operator; do not expire silently	MED-04, EMG-05
Concurrent medical and freight grants	Prioritize feasible medical new work without abandoning onboard trips	MED-01, FRT-02
Two SATSA candidates map to same (R,W)	Reuse cached fitness on same fixed traces; report unique candidate count	OPT-02/04

9.4 Quality Review Rules for the SRS

This student submission follows the requirement-quality guidance shown in Module 5 [10]. Each normative requirement should be necessary, singular, understandable, unambiguous in its simulator context, feasible at design level, consistent with subsystem authority and verifiable through a later observation or trace. Proposed parameter targets such as 1000 kg, the 10 s cumulative cap and the discrete SATSA search domain remain research and design baseline choices until the team signs them off.

Quality question	How the team checks it	Example
Is the actor or responsible module named?	Every "shall" belongs to Dispatch, Cabin, Control Center or Simulation Engine	REQ-04 names Dispatch reconciliation
Is the trigger clear?	Input/state/event is specified or referenced	**DOOR-04** triggers on obstruction during CLOSING
Is the result observable?	Trace field, UI transition or numerical oracle is identified	MED-04 generates UNAVAILABLE + alert
Is the model boundary explicit?	Simulator and physical safety are not conflated	MOVE-06 is not a real-world brake formula
Is the research claim bounded?	Cite the source and mark design adaptation separately	SATSA [8] does not report elevator waiting-time gains

9.5 Definition of Done and Design-Stage Approval

"SRS reviewed" and "Done" for an implemented user story are two different statuses. At submission, the team may meet the document-quality criteria while every implementation acceptance check is still NOT RUN. Later, a user story counts as implemented only after the code, the relevant unit and integration checks, acceptance evidence, error handling and traceability have been reviewed. That keeps the requirements document from presenting planned behavior as finished software.

- Design sign-off: the responsible team member has reviewed the scope, the unique Requirement ID, the acceptance condition, dependencies between subsystems and any open baseline decisions.
- Implementation Done (future): the behavior is built, its acceptance evidence is recorded, error branches are tested, no known critical contradiction remains and requirement links are up to date.
- Algorithm-study Done (future): test traces and seeds, baselines, the SATSA parameter domain, objective coefficients and the statistical report are archived. The paper's findings are not presented as this project's measurements.

9.6 System-Level Success Criteria and Research Traceability

A future prototype meets the project's intended system behavior when an operator can load the 40-floor/five-cabin baseline; create and deduplicate Hall Calls; see unique route ACK ownership; ride a simulated cabin to a destination;

inspect door and fault transitions; authorize MEDICAL and FREIGHT missions; reproduce input traces; and review SATSA recommendations, all without bypassing a Cabin safety check. We have to show this with recorded evidence at implementation time.

Research/engineering source	What it supports	What the Smart Elevator team proposes
[2] Siikonen (1997)	Elevator group modeling, dispatch and simulation context	Fleet-wide ETA and reproducible scenario framework
[3] Cortés et al. (2004)	Search-based strategies for elevator group control	Treat group assignment as an optimization/design problem
[4] Bolat & Cortés (2011)	Hall-call-to-car allocation optimization	Compare feasible multi-car assignment alternatives
[5] Crites & Barto (1998)	Alternative RL-based elevator group control research	Explain why a deterministic teaching prototype is chosen initially
[8] Liu et al. (2024)	SATSA mechanisms and constrained optimization evaluations	Apply SATSA offline to discrete (R,W) grouping tuning; test empirically later
[9] gRPC documentation	RPC service and Protocol Buffers communication concepts	Typed SubmitHallCall, ProposeRoute, ACK/NACK and telemetry contract
[10] Module 5 handout	Hybrid SRS, Agile backlog and traceability structure	Use stories and traceability alongside one system-level SRS

The team makes no quantitative claim that grouping improves elevator wait time, that SATSA beats grid search or that the proposed movement model is certified for physical lifts. We treat those as hypotheses for future acceptance and evaluation work.

A Glossary

Term	Definition
Hall Call	Public landing request identifying floor and UP or DOWN; destination unknown.
Cabin Destination	Floor chosen inside the cabin after boarding.
ETA	Estimated time of arrival at a planned stop from the modeled route.
Directional Grouping	Combining compatible same-direction calls within floor and time windows.
Commitment ETA	Reference arrival time for a stop when the system accepts it; not reset by optional future insertions.
ACK / NACK	Cabin acceptance or rejection of a versioned route proposal.
MEDICAL	Authorized medical transport mission prioritized against new work subject to safety.
FREIGHT	Authorized material transport mission with direct leg after passenger drain.
SAFE HANDOFF	Stationary, sensor-consistent and empty passenger cabin before exclusive service.
Canonical Hall Call	One active request per floor/direction, independent of number of button presses.
Route Version	Monotonically managed committed route revision used by ACK/NACK reconciliation.
Discrete Event	Time-stamped simulated event delivered by ordered event queue.
Definition of Done	Agile checklist for implemented behavior; not a claim that the SRS itself is coded.

B Open Decisions for Team Sign-off

Item	Proposed baseline	Team action
Simulation profile	40 floors; P1–P4 + G1; 2 s/floor; 6 s/stop	Confirm speed and dwell assumptions
Dispatch policy	R = 3, W = 5 s, DELAY_CAP = 10 s, AGE_LIMIT = 60 s	Approve or revise after scenario results
Capability policy	Which P-cars may serve MEDICAL / which material loads G1 permits	Approve per-car restrictions
Security implementation	Role-gated missions and traceable operator decisions	Select app authentication mechanism
SATSA offline research configuration	Proposed R=1..5 floors, W=0..10 s; reference (3,5); fixed λ and μ; separate training/held-out traces	Approve candidate domain, objective coefficients, evaluation budget and deployment process

Document status: instructor/team-review draft. The system requirements, state machines, user stories and test criteria are proposed implementation targets. Only the companion deterministic calculation checks have been run. SATSA optimization and gRPC transport are specified architectural features that still need integration evidence.

C Module 5 Coverage Crosswalk

This locator compares our hybrid SRS with the 29 categories in the instructor's Module 5 format-comparison handout [10]. It maps subject coverage, not matching heading numbers. The handout's 37-floor/six-cabin sample is teaching material. This project keeps its own 40-floor P1–P4/G1 baseline.

Module 5 area	SRS v1.1 section	Treatment
1 Introduction	1.1–1.5	Purpose, scope, definitions and cited literature
2 Overall Description	2.1–2.6	Context, product functions, actors, constraints
3 Stakeholders and Actors	2.3; 7.1	Passenger, staff, medical, operator roles
4 System Context	Figure 1; 2.1; 7.1	Three-subsystem boundary and adapters
5 Functional Requirements	4.1–4.8; 7	Normative DC/CB/CC/MED/FRT/SAF/OPT/etc.
6 Hall Calls	4.1; 7.2	Canonical deduplication and known fields
7 Dispatch	4.7; 7.3	Eligibility, ETA, grouping and route ownership
8 Movement	7.3	Continuous position, movement guards
9 Direction	4.7; 7.3	UP/DOWN/IDLE and safe direction behavior
10 Cabin interface	3.1; 7.8	Destination button, cabin and alarm feedback
11 Doors	7.4; Figure 8	Door state graph and obstruction
12 Safety	4.6; 5.2; 7.6	Cabin veto, faults, interlocks
13 Emergency	4.4; 7.6	MEDICAL distinct from ALARM/FAULT/FIRE
14 Sensors	3.2; 7.5	Position, door, load, obstruction, freshness
15 Simulation Engine	7.7; Figure 9	Clock, event queue, replay, fault injection
16 User Interface	3.1; 7.8	Hall, Cabin, Control Center, simulation views
17 Event Logging	7.9	Schema, example trace, replay data
18 Nonfunctional	5; 7.9	Performance, reliability, security, testability
19 Constraints	2.5; 4.7; 5.5	Hard constraints and simulator boundary
20 System States	7.2–7.6	Request, cabin, door and mission lifecycle
21 Agile Backlog	8.2–8.8	Epics E1–E7, US-001–US-016
22 Definition of Done	9.5	Design sign-off versus future implementation Done
23 Traceability	9.1	Requirement–story–AC–test matrix
24 Example Acceptance Test	9.2	Six representative scenarios and oracles
25 Edge Cases	9.3	Focused exceptional-case coverage
26 Agile Release Plan	8.9	Proposed staged increments R1–R5
27 Quality Rules	9.4	Necessary, consistent, observable, feasible
28 SRS ↔ Agile relationship	8.1; 9.1	One system specification maps to stories
29 Success Criteria	9.6	Future observable simulator objectives

This checklist records where each topic lives. It doesn't mean every future integration test has been run, and it doesn't claim the document follows any mandatory ISO/IEC/IEEE 29148 section numbering. Our approach is aligned with the handout's hybrid SRS guidance.

D Interface and Data Contract Examples

The contracts below let us describe gRPC/Protocol Buffers interoperability without tying the implementation to a particular language. They are conceptual payload examples, not a compiled .proto file. Each service boundary follows the ownership rules in Sections 3.4 and 7.2.

Logical contract	Required fields	Safety/consistency semantics
SubmitHallCall	origin_floor, direction, event_id, t_sim	Return canonical request_id; never infer destination
ProposeRoute	command_id, car_id, expected_route_version, ordered_stops	Cabin local feasibility check; versioned ACK or reasoned NACK
RouteAck	command_id, accepted, committed_version, reason	ACK confirms ownership; NACK preserves old route
CabinTelemetry	car_id, sim_time, position, direction, load, door_lock, mode, faults	Stale health values exclude eligibility
ApproveMission	mission_id, role, type, origin, destination, material_profile	Requires authorized Control Center grant and compatible cabin
OptimizationResult	run_id, theta_R, theta_W, objective, scenario_set, status	Offline recommendation; requires approval and safe activation

Idempotency invariant. Two messages with the same command_id and payload shall not commit two routes. A delayed ACK is resolved against the Cabin's actual accepted route version before a second owner can be confirmed. A repeated

command_id with a different payload is an error. Ordinary Hall Calls may merge by floor/direction without sharing personal identity.

Separation of concerns. Control Center can authorize a mission or policy version. Dispatch can rank and propose eligible routes. Only the Cabin Controller can accept or veto local route execution. SATSA outputs an offline configuration recommendation and is never a direct source of motor, door or emergency control commands.